



Lesson 1.6 — Function Transformations

Big idea: Every function transformation is one of four moves: shift, stretch/compress, reflect, or shift inside the parentheses. Learn the rules, then everything is plug-and-play.

Quick review

Outside the function (affects y): $f(x) + k$ shifts up k . $a \cdot f(x)$ stretches vertically by a (flips if $a < 0$).

Inside the function (affects x, and is BACKWARDS): $f(x - h)$ shifts right h . $f(bx)$ compresses horizontally by $1/b$ (flips if $b < 0$).

General form: $g(x) = a \cdot f(b(x - h)) + k$. Apply inside transforms first (h , b), then outside (a , k).

Sanity check: pick a known point on f , push it through the transformations, and verify it lands where g says it should.

Worked example

Worked example. Let $f(x) = x^2$. Describe and graph $g(x) = -2(x + 3)^2 + 1$.

Inside: $x + 3 = x - (-3) \rightarrow$ shift left 3. Outside: $\times 2$ stretches vertically by 2; the negative reflects across the x -axis; $+1$ shifts up 1.

Vertex moves from $(0, 0)$ to $(-3, 1)$; parabola opens DOWN, twice as steep.

Warm-ups

1. Describe how to get $g(x) = f(x) - 4$ from $f(x)$.
2. Describe how to get $g(x) = f(x - 5)$ from $f(x)$.
3. Describe how to get $g(x) = -f(x)$ from $f(x)$.

Practice

1. Let $f(x) = |x|$. Describe the transformations to get $g(x) = |x - 2| + 3$, and state the new vertex.
2. Let $f(x) = \sqrt{x}$. Describe the transformations to get $g(x) = 3\sqrt{x + 1} - 2$.
3. If $f(x)$ passes through $(2, 5)$, what point must $g(x) = f(x - 4) + 6$ pass through?

Challenge

1. Write a single equation $g(x)$ so that the parabola $y = x^2$ is reflected over the x -axis, stretched by a factor of 3, shifted right 4 and up 7. Then state its vertex and direction.
2. If $g(x) = f(-2x)$, describe what happens to the graph of f . Apply this to $f(x) = x^3 + x$ and find $g(1)$ and $g(-1)$.



Answer key — Lesson 1.6

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. Shift down 4.
2. Shift right 5.
3. Reflect across the x-axis (flip vertically).

Practice

1. Shift right 2 and up 3. New vertex: (2, 3).
2. Shift left 1, vertical stretch by 3, shift down 2.
3. (6, 11).

Challenge

1. $g(x) = -3(x - 4)^2 + 7$. Vertex (4, 7); opens DOWN.
2. Reflect across the y-axis, then horizontally compress by factor 1/2. With $f(x) = x^3 + x$: $g(1) = (-2)^3 + (-2) = -10$; $g(-1) = 2^3 + 2 = 10$.